Module 11

Quantitative Investment and Portfolio Management

November 10 & 12, 2025

Markowitz Theory ("Modern Portfolio Theory", MPT)

Mean and variance of a portfolio return:

Consider a single investment horizon T. Let $\{S_i\}$ = set of assets, $i = 1, 2, ..., n$. Let δS_i be the change of the value of the asset S_i for the period and $R_i = \frac{\delta S_i}{S_i}$ be its relative return (for simplicity assume S_i pays no dividend for now). Assume R_i follows normal distribution with mean $\mu_i T$, variance $\sigma_i^2 T$, and correlation coefficient ρ_{ij} for $i \neq j$. (μ_i = annualized expectation, σ_i^2 = annualized variance). For the moment assume $\sigma_i > 0$ for all i (i.e., there is no cash in the portfolio).

Consider a portfolio Π at $t = 0$

$\Pi = \sum_{i=1}^{n} w_i S_i$, (if $w_i < 0$ → short selling)

At T

$$\Pi \to \Pi + \delta\Pi = \sum_{i=1}^{n} w_i S_i (1 + R_i)$$

$$\frac{\delta\Pi}{\Pi} = \sum_{i=1}^{n} W_i R_i \quad , \quad W_i = \frac{w_i S_i}{\sum_{i=1}^{n} w_i S_i} \text{ (market value weight)}, \ \sum_{i=1}^{n} W_i = 1,$$

Thus for the portfolio:

$$\mu_\Pi = \frac{1}{T} E\left[\frac{\delta\Pi}{\Pi}\right] = \sum_{i=1}^{n} W_i \mu_i$$

$$\sigma_\Pi^2 = \frac{1}{T} Var\left[\frac{\delta\Pi}{\Pi}\right] = \sum_{i=1}^{n}\sum_{j=1}^{n} W_i W_j \sigma_{ij} = \sum_{i=1}^{n}\sum_{j=1}^{n} W_i W_j \rho_{ij} \sigma_i \sigma_j$$

$$\sigma_{ij} T = Cov(R_i, R_j) = E\left[(R_i - \mu_i T)(R_j - \mu_j T)\right]$$

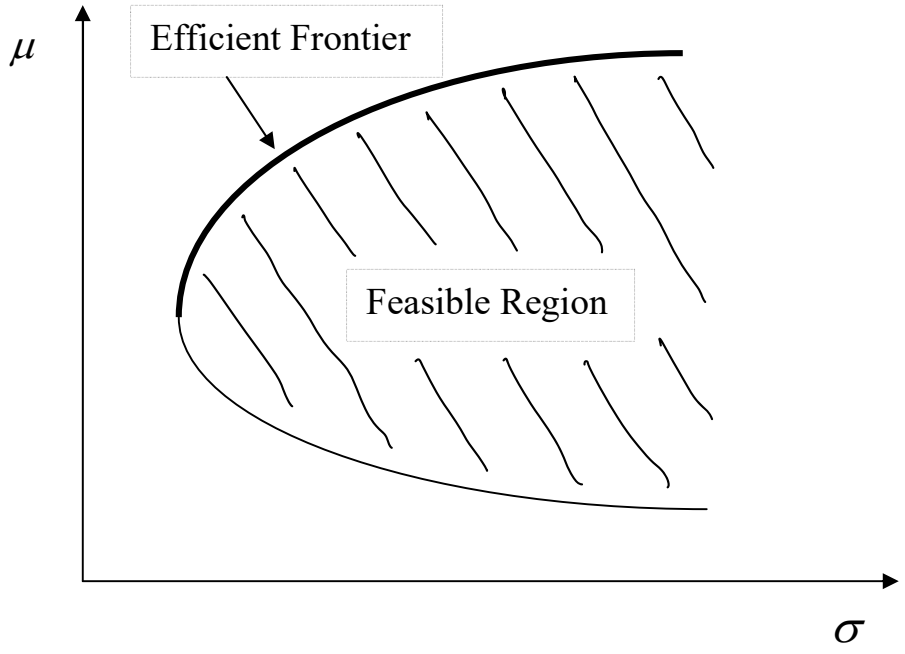
→ weight $\{\mu_i\},\{\sigma_{ij}\}$ by *market value* in computing μ_Π, σ_Π

Efficient portfolios:

Feasible region:

(σ, μ) for all the portfolios can be formed with the n assets:

$$\left\{(\sigma_\Pi, \mu_\Pi) \mid \{W_j\}, \sum_{i=1}^{n} W_i = 1\right\}$$



Efficient frontier:

$\{(\sigma_\Pi, \mu_\Pi)\}$ such that for a given σ_Π in the feasible region the portfolio has maximum μ_Π

Optimal investments:

- Given a risk tolerance σ → portfolio with maximum expected return μ
- Given an investment return target μ (in the upper feasible region) → portfolio with minimum variance σ^2

→ only portfolios on the efficient frontier should be selected

Markowitz problem:

Locating points on the efficient frontier:

Objective function:

$$\sum_{i=1}^{n}\sum_{j=1}^{n} W_i W_j \rho_{ij} \sigma_i \sigma_j$$

Constraint:

$$\sum_{i=1}^{n} W_i \mu_i = \tilde{\mu}$$

$$\sum_{i=1}^{n} W_i = 1$$

This is an optimization problem which can be solved via Lagrange multiplier:

$$L = \sum_{i=1}^{n}\sum_{j=1}^{n} W_i W_j \rho_{ij} \sigma_i \sigma_j - \lambda_1 \left(\sum_{i=1}^{n} W_i \mu_i - \tilde{\mu} \right) - \lambda_2 \left(\sum_{i=1}^{n} W_i - 1 \right)$$

$$\frac{\partial L}{\partial W_i} = 0 \,, \quad \frac{\partial L}{\partial \lambda_1} = 0 \,, \quad \frac{\partial L}{\partial \lambda_2} = 0$$

This yields $n+2$ equations with $n+2$ unknowns:

$$2\sum_{j=1}^{n} W_j \rho_{ij} \sigma_i \sigma_j - \lambda_1 \mu_i - \lambda_2 = 0 \ , \ i = 1, 2, ..., n$$

$$\sum_{i=1}^{n} W_i \mu_i - \tilde{\mu} = 0$$

$$\sum_{i=1}^{n} W_i - 1 = 0$$

A two asset portfolio example:

$$L = \left(W_1^2 \sigma_1^2 + 2 W_1 W_2 \rho_{12} \sigma_1 \sigma_2 + W_2^2 \sigma_2^2 \right) - \lambda_1 \left(W_1 \mu_1 + W_2 \mu_2 - \tilde{\mu} \right) - \lambda_2 \left(W_1 + W_2 - 1 \right)$$

$$\frac{\partial L}{\partial W_1} = \left(2 W_1 \sigma_1^2 + 2 W_2 \rho_{12} \sigma_1 \sigma_2 \right) - \lambda_1 \mu_1 - \lambda_2 = 0$$

$$\frac{\partial L}{\partial W_2} = \left(2W_2\sigma_2^2 + 2W_1\rho_{12}\sigma_1\sigma_2\right) - \lambda_1\mu_2 - \lambda_2 = 0$$

$$\frac{\partial L}{\partial \lambda_1} = W_1\mu_1 + W_2\mu_2 - \tilde{\mu} = 0$$

$$\frac{\partial L}{\partial \lambda_2} = W_1 + W_2 - 1 = 0$$

4 equations, 4 unknowns → $\{W_1, W_2\}$

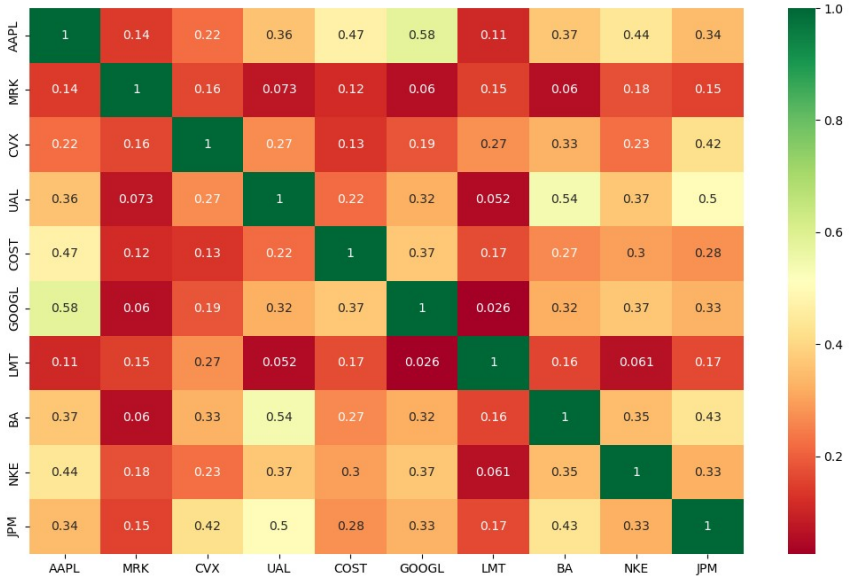
A numerical example of 10 asset portfolio:

Consider a 10 asset portfolio consists of shares in AAPL, MRK, CVX, UAL, COST, GOOGL, LMT, BA, NKE and JPM. Covariance matrix is estimated from the daily return during the past 5 years (11/09/2020 – 11/07/2025; data source: Yahoo Finance). The expected returns are estimated using the annualized mean returns in the same period.

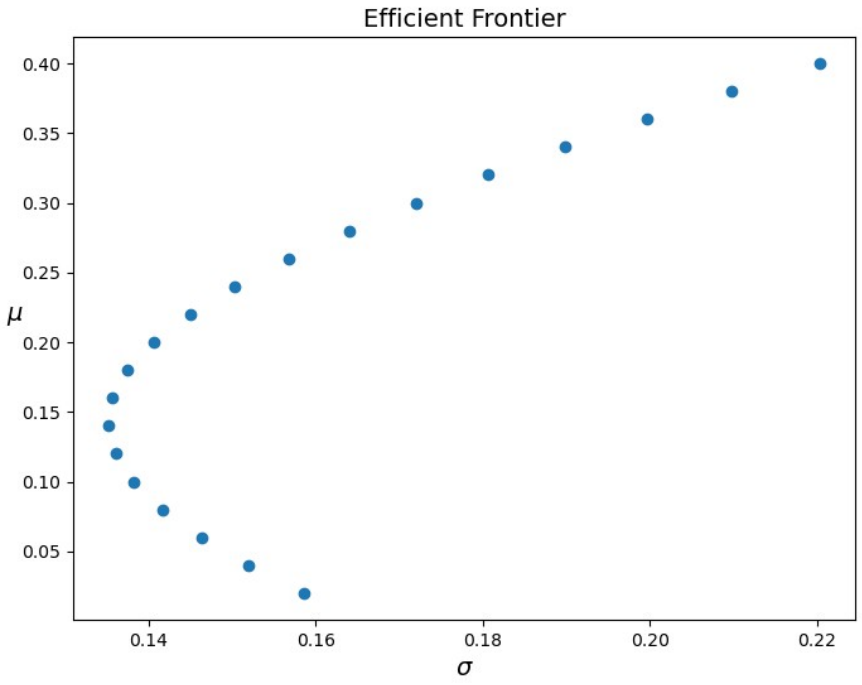
Annualized return and volatility:

Symbol	AAPL	MRK	CVX	UAL	COST	GOOGL	LMT	BA	NKE	JPM
Return	17.2%	5.5%	17.6%	17.1%	20.1%	23.1%	7.3%	1.6%	-13.6%	22.3%
Volatility	27.9%	22.6%	26.0%	47.9%	22.5%	30.7%	21.9%	37.2%	34.4%	24.2%

Correlation matrix:



The efficient frontier (for 1 year investment horizon) computed with the above procedure gives



The efficient frontier above is concave:

Let Π_A, Π_B be two portfolios with (μ_A, σ_A), (μ_B, σ_B) on the efficient frontier, then the combination

$$\Pi = x\Pi_A + (1-x)\Pi_B, \quad 0 < x < 1$$

is also a portfolio in the feasible region, with

$$\mu_\Pi = x\mu_A + (1-x)\mu_B, \quad \sigma_\Pi = \sqrt{x^2\sigma_A^2 + 2x(1-x)\rho_{AB}\sigma_A\sigma_B + (1-x)^2\sigma_B^2}$$

- $\rho_{AB} = 1$ → Π lies on the straight line in the (σ, μ) plane connecting Π_A and Π_B
- $\rho_{AB} < 1$ → Π moves left w.r.t. the $\rho_{AB} = 1$ point → above the connecting line

→ concavity

Merton: the efficient frontier above is a hyperbola ("An Analytic Derivation of the Efficient Portfolio Frontier", Journal of Financial and Quantitative Analysis, 7 (4), 1972, p1851-1872)).

Case of additional non-negative constraint (e.g., mutual funds):

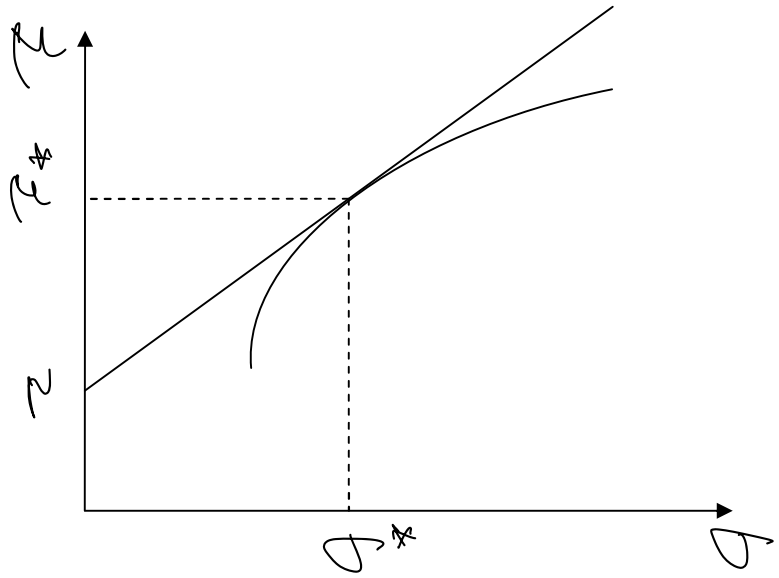
$W_j \geq 0$, $j = 1, 2, ..., n$

→ Quadratic programming

Combining with risk free asset B:

Assume interest rate on a risk free asset B is r, then $\mu_B = r$, $\sigma_B = 0$

Efficient frontier for (B, Π): $xB + (1-x)\Pi$



→ only *one* fund (σ^*, μ^*) is needed! (not the case if risk free asset B does not exist)

Remark: μ represents the *total* return (price appreciation + interest & dividend payout)

Practical issues with Markowitz portfolios:

- Difficulties with estimating expected returns
- Error in return estimation vs optimality in portfolio construction
- Large number of covariance to be estimated for a diversified portfolio

  → see later in the lecture when we discuss factor models
- Sensitivity to the off-diagonal elements of covariance matrix and their instability in real world

  → Hierarchical Risk Parity (HRP) approach to portfolio construction

Capital Asset Pricing Model (CAPM)

General assumptions:

- Assumptions on markets:
  - unique risk free asset, accessible to all investors
  - frictionless capital markets
- Assumptions on market participants (agents):
  - Portfolio construction is based on expected return and variance

    → justifiable if stock's return follows normal distribution

    (or investor's "utility" is quadratic)
  - all investors agree on distributions of security return and investment horizon

    → $\{\mu_i\}, \{\sigma_{ij}^2\}$ are the same for all investors

    → no "short term" or "long term" investors, they all on the same term

Some of the assumptions can be relaxed by considering, e.g., tax, transaction cost and heterogeneous expectations

Capital market line (CML):

What is the one fund that is all one needs?

→ everyone ends up at the same fund

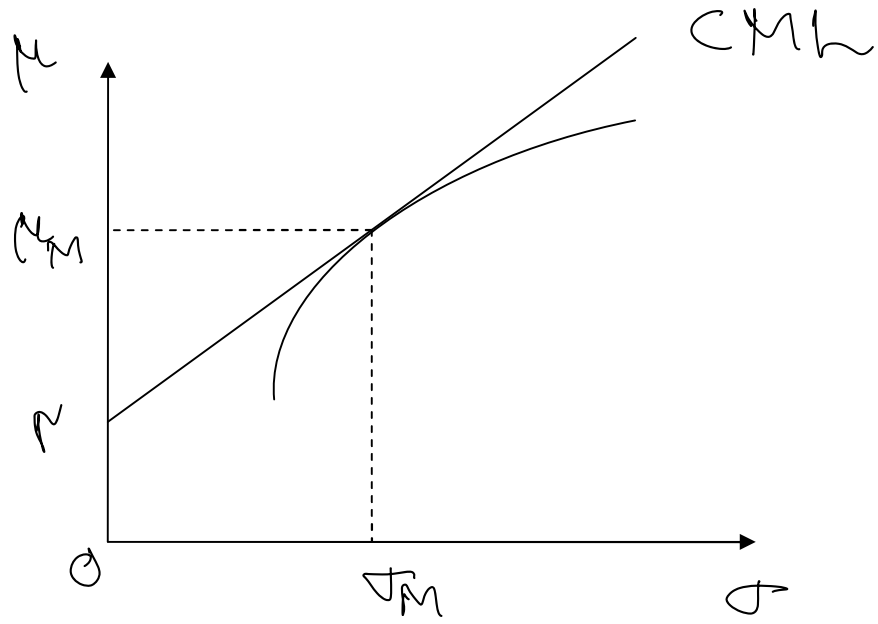
→ $\{\mu_i\}$, r will adjust such that this fund will be the entire market ("equilibrium")

→ market portfolio M

All market participants will construct their portfolio by combining the market portfolio and the risk free asset (B, M):

$\Pi = xB + (1-x)M$ ($x < 0$ → borrowing, etc)

→ capital market line on (σ, μ) plane:



$\mu_\Pi = r + \dfrac{\mu_M - r}{\sigma_M}\sigma_\Pi$, for all efficient portfolio Π.

Pricing for individual asset:

Expected return μ_i for asset S_i

$\mu_i = r + \beta_i\left(\mu_M - r\right)$,

$\beta_i = \dfrac{\sigma_{iM}}{\sigma_M^2} = \rho_{iM}\dfrac{\sigma_i}{\sigma_M}$ (beta of the asset S_i)

Proof:

Consider portfolio

$\Pi = xS_i + (1-x)M$

$\mu_\Pi = x\mu_i + (1-x)\mu_M$, $\sigma_\Pi = \sqrt{x^2\sigma_i^2 + 2x(1-x)\sigma_{iM} + (1-x)^2\sigma_M^2}$

At $x = 0$, Π = market portfolio

→ $\left.\dfrac{d\mu_\Pi}{d\sigma_\Pi}\right|_{x=0}$ must be the same as the slope of CML (otherwise CML will not be efficient)

$\dfrac{d\mu_\Pi}{dx} = \mu_i - \mu_M$,

$$\frac{d\sigma_\Pi}{dx} = \frac{x\sigma_i^2 + (1-2x)\sigma_{iM} - (1-x)\sigma_M^2}{\sigma_\Pi} \rightarrow \left.\frac{d\sigma_\Pi}{dx}\right|_{x=0} = \frac{\sigma_{iM} - \sigma_M^2}{\sigma_M}$$

$$\left.\frac{d\mu_\Pi}{d\sigma_\Pi}\right|_{x=0} = \left.\frac{\frac{d\mu_\Pi}{dx}}{\frac{d\sigma_\Pi}{dx}}\right|_{x=0} = \frac{(\mu_i - \mu_M)\sigma_M}{\sigma_{iM} - \sigma_M^2} = \frac{\mu_M - r}{\sigma_M} \text{ (CML)}$$

$$\rightarrow \mu_i = r + \frac{\sigma_{iM}}{\sigma_M^2}(\mu_M - r) = r + \beta_i(\mu_M - r) \qquad \text{Q.E.D.}$$

CAPM links expected return of an individual stock to that of the market via β

$\rightarrow \beta_i$ (together with μ_M) is all one needs to know about expected return in S_i

Example:

β of the 10 stocks in the portfolio considered above estimated using the data sampled in the same period against SP500 as a proxy for the market which had a mean annualized return 12.74%.

Symbol	AAPL	MRK	CVX	UAL	COST	GOOGL	LMT	BA	NKE	JPM
beta	1.250	0.288	0.632	1.563	0.778	1.259	0.260	1.183	1.107	0.897

What about R_i, the actual rate of return of S_i?

Under CAPM

$$R_i = r + \beta_i(R_M - r) + \varepsilon_i, \quad E(\varepsilon_i) = 0$$

$$Cov(R_i, R_M) = \beta_i\sigma_M^2 + Cov(\varepsilon_i, R_M) \rightarrow Cov(\varepsilon_i, R_M) = 0$$

$$\rightarrow \sigma_i^2 = \beta_i^2\sigma_M^2 + Var(\varepsilon_i)$$

Furthermore, ε_i's are not correlate with each other, i.e., $E(\varepsilon_i\varepsilon_j) = 0$, $i \neq j$: assume the market contains N assets, if $E(\varepsilon_i\varepsilon_j) = 0$

$$\sigma_M^2 = \sum_{i=1}^N\sum_{j=1}^N W_iW_j\sigma_{ij} = \sum_{i=1}^N\sum_{j=1}^N W_iW_j\beta_i\beta_j\sigma_M^2 = \sigma_M^2\left(\sum_{i=1}^N W_i\beta_i\right)^2 = \sigma_M^2\beta_M^2 = \sigma_M^2 \rightarrow \text{ consistent!}$$

- $\beta_i^2\sigma_M^2$: systematic risk
- $Var(\varepsilon_i)$: idiosyncratic risk, firm specific

Why idiosyncratic risk $Var(\varepsilon_i)$ does not contribute to the expected return μ_i? Where is the reward for taking such risk?

- Idiosyncratic risk is uncorrelated with market and with each other, can be diversified away in a large portfolio (more later when we discuss factor model)

  → no "risk premium": if $Var(\varepsilon_i) > Var(\varepsilon_j)$ but $\beta_i = \beta_j$, one still has $\mu_i = \mu_j$

- Systematic risk cannot be eliminated by diversification

Extreme cases:

- $\beta_i = 0$: $\mu_i = r$? → risk can be diversified away (i.e., no reward for taking diversifiable risk)
- $\beta_i = -1$: $\mu_i < r$? → contrarian view, short candidates; risk reduction in portfolio construction

Discount rate for risky asset S_i :

$$\mu_i T = E(R_i) = E(\frac{\delta S_i}{S_i}) = \frac{E(S_i^T) - S_i}{S_i} = \left(r + \beta_i \left(\mu_M - r\right)\right)T$$

$$\rightarrow \ S_i = \frac{E(S_i^T)}{1 + \left(r + \beta_i \left(\mu_M - r\right)\right)T}$$

$r + \beta_i \left(\mu_M - r\right)$ = risky discount rate for S_i → determined by its beta β_i

Linearity of the pricing formula:

Let $\Pi_{i,j} = S_i + S_j$

$$\mu_{i,j} T = \frac{E(S_i^T + S_j^T) - (S_i + S_j)}{S_i + S_j} = \left(r + \beta_{i,j} \left(\mu_M - r\right)\right)T$$

$$\rightarrow \ S_i + S_j = \frac{E(S_i^T + S_j^T)}{1 + \left(r + \beta_{i,j} \left(\mu_M - r\right)\right)T}, \quad \beta_{i,j} = \frac{\sigma_{\Pi_{i,j} M}}{\sigma_M^2}$$

Is this the same as

$$S_i + S_j = \frac{E(S_i^T)}{1 + \left(r + \beta_i \left(\mu_M - r\right)\right)T} + \frac{E(S_j^T)}{1 + \left(r + \beta_j \left(\mu_M - r\right)\right)T} \ ?$$

Not obvious, but yes:

$$\beta_i = \frac{\sigma_{iM}}{\sigma_M^2} = \frac{Cov(R_i, R_M)}{\sigma_M^2 T} = \frac{Cov(\frac{S_i^T}{S_i} - 1, R_M)}{\sigma_M^2 T} = \frac{Cov(S_i^T, R_M)}{S_i \sigma_M^2 T}$$

$$\rightarrow\ S_i = \frac{E(S_i^T)}{1 + \left(r + \beta_i \left(\mu_M - r\right)\right)T} = \frac{E(S_i^T)}{1 + \left(r + \frac{Cov(S_i^T, R_M)}{S_i \sigma_M^2 T}\left(\mu_M - r\right)\right)T}$$

$$\rightarrow\ S_i = \frac{1}{1+rT}\left(E(S_i^T) - \frac{Cov(S_i^T, R_M)(\mu_M - r)}{\sigma_M^2}\right) \rightarrow \text{ linearity in } S$$

$$\rightarrow \left(E(S_i^T) - \frac{Cov(S_i^T, R_M)(\mu_M - r)}{\sigma_M^2}\right) \text{ is discounted as if risk free}$$

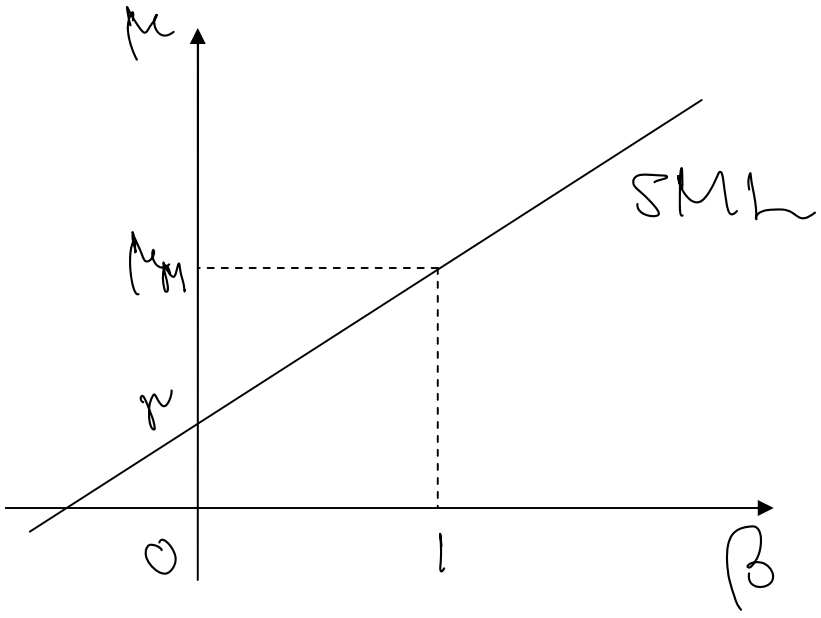
→ "certainty equivalent"

<u>Security market line (SML)</u>:

CAPM indicates that the expected return of assets is a linear function of its β:

$$\mu_i = r + \left(\mu_M - r\right)\beta_i$$

On the (β, μ) plane:



- $\beta > \beta_M = 1$: aggressive, $\beta < 1$: defensive
- Points above SML: under-priced assets; below SML: over-priced assets

Impact of CAPM on investment practice:

- Popularity of passive (index) investing

  Criticism: fewer people scrutinizing company details

  → less efficient markets

- Popularity of market capitalization weighting in index construction

  Criticism: it may lead to:

  - large concentrations of a few assets in an index
  - high volatility due to concentration and herd behavior
  - inferior performance due to over weighting overvalued assets and under weighting undervalued assets

Performance measurements:

- "Active portfolio" → "alpha": excess return over its benchmark
- Risk adjusted return → "Sharpe ratio" :
  - CML: $\frac{\mu_\Pi - r}{\sigma_\Pi} = \frac{\mu_M - r}{\sigma_M}$
  - Active equity portfolio: strive to achieve
    - $R_\Pi > R_M$
    - $\frac{\mu_\Pi - r}{\sigma_\Pi} > \frac{\mu_M - r}{\sigma_M}$
- A curious statistics:

  "Over the past 20 years, only 17.6% of active U.S. equity funds beat the S&P 500. Meanwhile, active bond funds outperformed their benchmark 59% of the time." (source: Morningstar, July 2025)

  Possible explanations: ….

Arbitrage Pricing Theory (APT)

Linear factor model:

One period return of asset S_i follows a linear factor structure (for simplicity assume the period $T = 1$ hence no need of annualization of μ_i etc):

$$R_i = \mu_i + \sum_{l=1}^{k} \beta_{il} f_l + \varepsilon_i \ , \ \ i = 1,2,...,n \ , \ l = 1,2,...,k$$

f_l = factors: $E(f_l) = 0$, $E(f_l f_m) = \sigma_{lm}^f$

- factors may not be identifiable
- possible factors:
  - Market return R_M
  - GDP
  - Treasury yield
  - Inflation
  - ….
- $E(f_l) = 0$ per definition: if not, one can redefine f_l and absorb the non-zero mean into μ_i :

  $$f_l \to f_l' = f_l - E(f_l) \ , \quad \mu_i \to \mu_i' = \mu_i + \beta_{il} E(f_l)$$

β_{il} = "factor loading" of S_i to the factor f_l (also called "factor beta")

ε_i = noise (idiosyncratic risk), uncorrelated with factor returns and with each other:

$$\varepsilon_i \sim N(0, e_i^2) \ , \ E(\varepsilon_i \varepsilon_j) = 0 \ , \ i \neq j \ ; \ E(f_l \varepsilon_i) = 0 \ , \ \forall i,l$$

→ $\mu_i = E(R_i)$

APT:

In an arbitrage-free world with many assets and a linear factor structure for rates of return, the expected rate of return of any asset is a linear function of its sensitivities to the factor risks

Proof:

Assume assets $\{S_1, S_2, ..., S_n\}$ with $n \gg k$

$$\begin{pmatrix} R_1 \\ R_2 \\ \vdots \\ R_n \end{pmatrix} = \begin{pmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_n \end{pmatrix} + \begin{pmatrix} \beta_{11} & \beta_{12} & \cdots & \beta_{1k} \\ \beta_{21} & \beta_{22} & \cdots & \beta_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ \beta_{n1} & \beta_{n2} & \cdots & \beta_{nk} \end{pmatrix} \begin{pmatrix} f_1 \\ f_2 \\ \vdots \\ f_k \end{pmatrix} + \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix} \quad \Leftrightarrow \quad \mathbf{R} = \boldsymbol{\mu} + \mathbf{B}\mathbf{f} + \boldsymbol{\mathcal{E}}$$

Consider a zero beta portfolio:

$$\mathbf{P}^{\mathbf{T}}\mathbf{B} = \mathbf{0}, \quad \begin{pmatrix} w_1 S_1 \\ w_2 S_2 \\ \vdots \\ w_n S_n \end{pmatrix} \equiv \begin{pmatrix} P_1 \\ P_2 \\ \vdots \\ P_n \end{pmatrix} \equiv \mathbf{P}, \quad \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \equiv \mathbf{0}, \quad \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix} \equiv \mathbf{1} \quad \Leftrightarrow \quad \sum_{i=1}^{n} w_i S_i \beta_{il} = 0, \; l = 1, 2, ..., k$$

To construct such a portfolio one needs to solve k equations with n unknown:

$n \gg k$ → infinite many such portfolios (disregard the trivial solution $\mathbf{P} = \mathbf{0}$)

→ exist many well diversified zero beta portfolios where $W_i \sim \frac{1}{n}$

→ for such portfolios, $\sum_{i=1}^{n} W_i \varepsilon_i \to 0$, $n \to \infty$

→ portfolio return = $\mathbf{P}^{\mathbf{T}}\mathbf{R} = \mathbf{P}^{\mathbf{T}}\left(\boldsymbol{\mu} + \mathbf{B}\mathbf{f} + \boldsymbol{\mathcal{E}}\right) = \mathbf{P}^{\mathbf{T}}\boldsymbol{\mu}$

No arbitrage → $\mathbf{P}^{\mathbf{T}}\boldsymbol{\mu} = r$ (return of a position without uncertainty) → $\mathbf{P}^{\mathbf{T}}\left(\boldsymbol{\mu} - r\mathbf{1}\right) = 0$

Linear algebra: $\forall \mathbf{u}$, $\mathbf{u} \perp \{\mathbf{v}_1, \mathbf{v}_2, ..., \mathbf{v}_m\} \to \mathbf{u} \perp \mathbf{v}_{m+1}$, then $\mathbf{v}_{m+1} = \sum_{i=1}^{m} c_i \mathbf{v}_i$ (exercise)

$$\mathbf{P}^{\mathbf{T}}\mathbf{B} = \mathbf{0} \to \mathbf{P}^{\mathbf{T}}\left(\boldsymbol{\mu} - r\mathbf{1}\right) = 0 \quad \rightarrow \quad \boldsymbol{\mu} = r\mathbf{1} + \mathbf{B}\boldsymbol{\lambda}, \; \boldsymbol{\lambda} \equiv \begin{pmatrix} \lambda_1 \\ \lambda_2 \\ \vdots \\ \lambda_k \end{pmatrix}$$

→ $E(R_i) = \mu_i = r + \sum_{l=1}^{k} \lambda_l \beta_{il}$ → asset return is determined by factor loadings

→ $R_i = r + \sum_{l=1}^{k} \beta_{il}(\lambda_l + f_l) + \varepsilon_i$

Meaning of λ_l:

Suppose factors are independent and the asset S_j with $\beta_{jl} = 1$, $\beta_{jl'} = 0$, $l \neq l'$

→ $E(R_j) = \mu_j = r + \lambda_l$

→ λ_l = risk premium ("price of risk") of the factor f_l

Security Market (Hyper-) Plane under APT:

$$\mu_i = r + \sum_{l=1}^{k} \lambda_l \beta_{il}$$

Expected return of an asset lies above or below this hyper-plane → under or over priced

Arbitrage under APT:

As established above the firm specific risk $\mathcal{E}$ cancels out in a well diversified portfolio so that its return, $\mathbf{P^T R} = \mathbf{P^T}\left(\boldsymbol{\mu} + \mathbf{Bf}\right)$, is solely determined by its factor exposure

→ Portfolios with identical factor exposure must have the same expected return

→Arbitrage opportunity arises if it is not the case

<u>Advantage over CAPM</u>:

- no need to assume equilibrium
- may incorporate more factors

<u>Critique over APT</u>:

- no consensus on what factors should be
- not uniquely defined (once factors identified, there are infinite many linear combinations are equally valid)
- in practice there never be infinitely many assets hence idiosyncratic risk can never be truly diversified away
- hence APT is not refutable or falsifiable

Factor Models

Single factor model:

Assume the market return R_M embeds all the information pertinent to forecast asset return R_i one may adopt a single factor model where the factor $\tilde{R}_M = R_M - \mu_M$

→ looks similar to CAPM but not equivalent:

- CAPM: it is sufficient to estimate the expected return of an asset S_i from its β_i
- Factor Model: not an equilibrium model, one needs to compute μ_i separately

Risk analysis of a diversified portfolio under a single factor model:

Assume

$R_i = \mu_i + \beta_i \tilde{R}_M + \varepsilon_i$, $\varepsilon_i \sim N(0, e_i^2)$, $E(\varepsilon_i \tilde{R}_M) = 0$, $E(\varepsilon_i \varepsilon_j) = 0$, $i \neq j$

→ $\sigma_i^2 = \beta_i^2 \sigma_M^2 + e_i^2$

→ $R_\Pi = \dfrac{\delta \Pi}{\Pi} = \sum_{i=1}^{n} W_i R_i = \sum_{i=1}^{n} W_i \left(\mu_i + \beta_i \tilde{R}_M + \varepsilon_i \right)$

→ $\mu_\Pi = \sum_{i=1}^{n} W_i \mu_i$,

$\sigma_\Pi^2 = \sum_{i=1}^{n} \sum_{j=1}^{n} W_i W_j \beta_i \beta_j \sigma_M^2 + \sum_{i=1}^{n} W_i^2 e_i^2$

Well diversified portfolio → $W_i \sim \dfrac{1}{n}$ (no single asset dominates the portfolio)

→ $\sigma_\Pi^2 \sim \sum_{i=1}^{n} \sum_{j=1}^{n} W_i W_j \beta_i \beta_j \sigma_M^2 = \sigma_M^2 \left(\sum_{i=1}^{n} W_i \beta_i \right)^2 = \sigma_M^2 \beta_\Pi^2$, $n \to \infty$ $\quad (\beta_\Pi = \sum_{i=1}^{n} W_i \beta_i)$

→ idiosyncratic risk diversified away

→ only risk is the "systematic risk", i.e., the risk associated with the Market

These conclusions can be generalized to multi-factor cases:

- systematic risk: risk associated with factors gives the risk premium of a portfolio
- idiosyncratic risk: can be diversified away

Multi-factor models in mean-variance optimization:

- Markowitz Theory:

  # of parameters need be estimated in a mean-variance optimization for a portfolio of n assets:

  n (expected return) + n (variance) + n(n – 1)/2 (covariance)

  = 2n + n(n – 1)/2

  - n = 500: # of parameters = 125,750
  - n = 2000: # of parameters = 2,003,000

- Factor Model:

  # of parameters need be estimated in a k-factor model:

  n (expected return) + nk (factor loading) + k (factor variance) +

  k(k – 1)/2 (factor covariance) + n (asset idiosyncratic variance)

  = 2n + nk + k + k(k – 1)/2

  - n = 500: k = 3, # of parameters = 2,506; k = 10, # of parameters = 6,055
  - n = 2000: k = 3, # of parameters = 10,006; k = 10, # of parameters = 24,055

  → Substantial reduction in computational cost

Statistical estimate of covariance matrix for factors are in general also more stable than that for individual assets

Factor selection:

314 empirical factors listed in "… and the Cross-Section of Expected Returns", Campbell R. Harvey, Yan Liu, Heqing Zhu, The Review of Financial Studies, Volume 29, Issue 1, January 2016, Pages 5–68, https://doi.org/10.1093/rfs/hhv059

→ "factor zoo"

- External (macroeconomic) factors:
  - GDP, CPI, PMI, …
  - yield curve level, yield curve shape (steepness), …
  - credit spread level
  - …

- Fundamental factors:
  - Price-to-earnings ratio (P/E)
  - Price-to-book value (P/B)
  - Price-to-sales ratio (P/S)
  - Price-to-cash flow ratio (P/CF),
  - Price/earnings to growth ratio (PEG ratio),
  - Debt-to-Equity Ratio (D/E)
  - …
- Extracted (statistical) factors:
  - Total market return
  - Sector return
  - …
- Technical factors:
  - Liquidity (bid/ask spread, volume, etc)
  - Momentum (e.g., year-over-year growth)
  - Sentiment
  - …

Recent developments:

- "Factor Models, Machine Learning, and Asset Pricing", Stefano Giglio, Bryan Kelly and Dacheng Xiu, Annual Review of Financial Economics Volume 14, 2022.
- Open Source Asset Pricing: https://www.openassetpricing.com/ (Andrew Chen and Tom Zimmermann, last updated October 2025)
- "APT or 'AIP'? The Surprising Dominance of Large Factor Models", A. Didisheim et al, 2025, https://www.nber.org/papers/w33012
- Factor mining from "alternative data":
  - Individual data: social media (X, LinkedIn, …), news & reviews, web search, …
  - Transaction data: consumers, corporations, government agencies

    - ESG (Environmental, Social, Governance) scores
    - Sensor data: satellite images, geolocations (GPS, apps, …) , weather, …
- Machine learning: synthetic factors identified via, e.g., PCA, ML feature selections (from "latent space"), …

Data and analytics vendors:

- MSCI Barra (Morgan Stanley Capital International Barra)
- …

Factor models in practice:

- Separate factor models are often used for predicting excess return (alpha) and risk (beta)
- Factors affecting short term and long term return may be significantly different, the statistical estimation over different time horizon may also be different, and thus the signals captured in the models may be different.
- Factor model selection:
    - bottom up investment strategy: fundamental models
    - top down investment strategy: macroeconomic models
    - importance of economic/financial intuition
- Data issues with implementing factor models:
    - model alignment (e.g., accounting methods used to compute fundamental factors)
    - survival bias
    - look ahead bias
    - data snooping (or data fishing or data dredging)

Factor Investment Strategies:

- "Smart beta"
- Factors selection → finding alpha

- Sources of excess return get identified and utilized by quantitative models:

  beta → alpha.

- Allocating investments to selected factors rather than asset classes

  → avoid concentrations of risk that result from overlap in the factor exposures of asset classes

- Factor-mimicking portfolio (FMP)
- …

Portfolio optimizations in practice:

- General formulation:
  - decision variables
  - objective function
  - constraints
- Constraint:
  - Long only constraint
  - Factor constraints (limit on factor exposure)
  - Policy (e.g., ESG) constraints
  - Holding size constraints (cap/floor on individual asset in the portfolio)
  - Cardinality constraints (cap on the # of assets in the portfolio)
  - Turnover constraints (cap on trading frequency)
  - Round lots constraints
  - Tracking error constraints
- "hard" vs "soft" constraints